

Generation Interconnection System Impact Study Report

For Queue Project Q16-0694

Thomas Hill Station 500 kV

January 24, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0694. The Interconnection Customer proposes a 384.9 MW solar photovoltaic facility interconnecting at the Thomas Hill Station 500 kV substation in the Indiana area. The request was received on October 24, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 306 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,375,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Villa Park to Washington County Cogeneration Facility 230 KV
Contingency: Barrett to Wolfe Substation 500 KV
Loading: 114.2% (Rating: 856 MVA)
DFAX: 5.1%
Pre-project loading: 111.9%
2. Facility: Villa Park to Wolfe 230 KV
Contingency: Wolfe to Barrett 500 KV line
Loading: 109.8% (Rating: 920 MVA)
DFAX: 5.1%
Pre-project loading: 107.7%
3. Facility: Imperial to Chamberlin 230 KV
Contingency: Ravenna to Hyleos 500 KV
Loading: 108.8% (Rating: 563 MVA)
DFAX: 9.4%
Pre-project loading: 102.3%
4. Facility: Allegheny Energy Units 3 4 & 5 500/1kV XFMR
Contingency: Piney Crk Project to Hilltop 500 KV
Loading: 108.6% (Rating: 1,302 MVA)
DFAX: 9.4%
Pre-project loading: 105.8%
5. Facility: Allegheny Energy Units 3 4 & 5 345/1 kV Autotransformer
Contingency: Hilltop to Piney Crk Project Substation 500 KV
Loading: 108.6% (Rating: 1,302 MVA)
DFAX: 9.4%
Pre-project loading: 105.8%
6. Facility: Scott to Shawville Station 345 KV
Contingency: Orchard Rd to Shawville Sta 500 KV
Loading: 108.2% (Rating: 1,401 MVA)

DFAX: 5.0%

Pre-project loading: 106.8%

7. Facility: Sugar Hill to Allegheny Energy Units 3 4 & 5 345 KV

Contingency: Hilltop to Piney Crk Project 500 KV

Loading: 108.0% (Rating: 1,438 MVA)

DFAX: 9.7%

Pre-project loading: 105.4%

8. Facility: Piney Creek Roject to Handsome Lake 500 KV L

Contingency: Lions Rd to Piny Creek Project 500 KV Line

Loading: 104.7% (Rating: 2,282 MVA)

DFAX: 11.8%

Pre-project loading: 102.7%

9. Facility: Caldwell to Piney Crk Project 345 KV line

Contingency: Hilltop to Piney Crk Project 500 KV

Loading: 103.2% (Rating: 1,917 MVA)

DFAX: 8.5%

Pre-project loading: 101.5%

10. Facility: P.T.S to Medora 345 KV Line

Contingency: Allegheny Energy Units 3 4 & 5 to Sugar Hill 345 KV line

Loading: 100.8% (Rating: 1,619 MVA)

DFAX: 7.7%

Pre-project loading: 99.0%

11. Facility: Acadia to Montour Sta 500 KV

Contingency: Coronet to Westwood Generation Llc 500 KV

Loading: 100.5% (Rating: 2,796 MVA)

DFAX: 8.1%

Pre-project loading: 99.4%

12. Facility: Thomas Hill Station 500-230 kV Transformer Bank

Contingency: Montpelier Electric Generating Station to Wabash 500 KV

Loading: 100.2% (Rating: 1,009 MVA)

DFAX: 22.7%

Pre-project loading: 91.5%

13. Facility: Ravenna to Old Forge 230 KV

Contingency: Ravenna to Hylebos 500 KV

Loading: 100.1% (Rating: 725 MVA)

DFAX: 8.1%

Pre-project loading: 95.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Villa Park to Washington County Cogeneration Facility 230 KV	Rebuild line with higher-capacity conductor	\$25,523,000	4% (\$1,084,000)
Villa Park to Wolfe 230 KV	Reconductor line	\$5,851,000	4% (\$249,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Imperial to Chamberlin 230 KV	Reconductor line	\$4,559,000	4% (\$171,000)
Allegheny Energy Units 3 4 & 5 500/1 kV Autotransformer	Replace transformer with higher-rated unit	\$45,349,000	4% (\$1,592,000)
Allegheny Energy Units 3 4 & 5 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$14,271,000	4% (\$501,000)
Scott to Shawville Sta 345 KV	Reconductor line	\$25,874,000	8% (\$2,128,000)
Sugar Hill to Allegheny Energy Units 3 4 & 5 345 KV	Reconductor line	\$7,114,000	4% (\$250,000)
Handsome Lake to Piney Crk Project 500 KV	Reconductor line	\$35,796,000	4% (\$1,252,000)
Piney Creek Project to Caldwell Substation 345 KV	Reconductor line	\$26,047,000	4% (\$913,000)
Medora to P.T.S 345 KV L	Reconductor line	\$12,528,000	4% (\$437,000)
Acadia to Montour Sta 500 KV	Reconductor line	\$46,577,000	4% (\$1,625,000)
Thomas Hill Station 500-230 kV Autotransformer	Replace transformer with higher-rated unit	\$26,453,000	15% (\$3,977,000)
Old Forge to Ravenna 230 KV	Reconductor line	\$5,599,000	4% (\$196,000)
	Total allocated to project		\$14,375,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.